ONLINE ASSESSMENT · 10 OCT 2026

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Campus: IIT Delhi

1. Questions

Pattern: 3 coding questions

Q1. Minimize Tree Height by Re-attaching Edges Coding

You are given a tree with n vertices rooted at vertex 1, and an integer `max_operations`.

- One **operation**: choose an edge (v, u), where v is the parent of u, remove it, and connect u (with its whole subtree) **directly to the root**.
- You may perform **at most** `max_operations` operations.
- The **height** of the tree is the maximum depth of any vertex (depth = number of edges from the root).

Return: the minimum possible height of the tree.

> Practice (same problem): Codeforces 1739D — Reset K Edges

Q2. Maximum Profit with Bitwise OR at Most K Coding

You are given two arrays `a` and `profits` of length n, and an integer k. Item i has number `a[i]` and profit `profits[i]`.

- Choose any subset of items such that the **bitwise OR** of their `a` values is **at most** k.
- The profit of a subset is the sum of `profits[i]` over the chosen items.

Return: the maximum total profit.

> Practice (same problem): AtCoder Tenka1 2017 D — IntegerotS

Q3. Subsequence With the Minimum Score Coding

You are given two strings `one` and `two`. You may remove any number of characters from `two`.

- After the removals, `two` must be a **subsequence** of `one`.
- **Score**: if no character is removed, the score is 0. Otherwise, let first and last be the smallest and largest indices (in the original `two`) of the removed characters; the score is $\text{last} - \text{first} + 1$.

Return: the minimum possible score.

> Practice (same problem): LeetCode 2565 — Subsequence With the Minimum Score